

Numerical Methods 1

1 c

$$2 \quad 4n-2$$

$$-46$$

$$-46$$

$$- \frac{46}{2} (2+2 + (46-1) \times 4)$$

$$= 46(1+90)$$

$$= 46 \cdot 91$$

$$= \underline{4186}$$

$$3 \quad y = x^{\frac{1}{2}} \cdot (2-x)^3$$

$$y' = x^{\frac{1}{2}} \cdot -3(2-x)^2 + \left(\frac{1}{2} x^{-\frac{1}{2}} \cdot (2-x)^3 \right)$$

$$= -3\sqrt{x}(2-x)^2 + \frac{1}{2\sqrt{x}}(2-x)^3$$

$$y'(4) = -3\sqrt{4} \cdot (2-4)^2 + \frac{1}{2\sqrt{4}}(2-4)^3$$

$$= -6 \cdot 4 + \frac{1}{4} \cdot -8$$

$$= -24 - 2$$

$$= -26$$

$$4 \quad y = (1-x)(1+2x)^{-\frac{1}{2}}$$

$$1 + \left(\frac{1}{2} \cdot 2x \right) + \left(\frac{-\frac{1}{2} \cdot -\frac{3}{2}}{2!} (2x)^2 \right)$$

$$= 1 - x - \frac{3}{2}x^2$$

$$\text{coefficient of } x = -1$$

$$\text{of } x^2 = -\frac{3}{2}$$

$$|x| < \frac{1}{2}$$

$$5 \quad x = 2t - 3$$

$$\frac{dx}{dt} = 2$$

$$y = t^2 - 2t^2$$

$$\frac{dy}{dt} = 2t^2 - 2t$$

$$\frac{dy}{dx} = \frac{2t^2 - 2t}{2}$$

$$= \frac{3}{2}t^2 - t$$

$$\frac{3}{2}t^2 - t = 0$$

$$\rightarrow 3t^2 - 2t = 0$$

$$t(3t-2) = 0$$

$$t = \frac{2}{3} \text{ or } 0$$

$$\text{At } t = \frac{2}{3}, \quad x = -\frac{5}{3}$$

$$y = -\frac{2}{27}$$

Consolidation

$$1 \begin{array}{c|c} x & 0, 0.5, 1, 1.5, 2 \\ y & 1, \frac{4}{5}, \frac{1}{2}, \frac{4}{13}, \frac{1}{5} \end{array}$$

$$h = 0.5$$

$$\int_0^2 \frac{1}{1+x^2} dx \approx \frac{0.5}{2} \left[1 + 2 \left(\frac{4}{5} + \frac{1}{2} + \frac{4}{13} \right) + \frac{1}{5} \right] \\ \approx 1.104$$

$$2 \begin{array}{c|c} x & 0 \quad 0.5 \quad 1 \quad 1.5 \quad 2 \\ y & 1 \quad \frac{3\sqrt{2}}{4} \quad \sqrt{2} \quad \frac{\sqrt{10}}{4} \quad 3 \end{array}$$

$$\text{under: } 0.5 \times \left(1 + \frac{3\sqrt{2}}{4} + \sqrt{2} + \frac{\sqrt{10}}{4} \right) \\ = 2.783 \quad \text{under by } 0.458$$

$$\text{over: } 0.5 \times \left(\frac{3\sqrt{2}}{4} + \sqrt{2} + \frac{\sqrt{10}}{4} + 3 \right) \\ = 3.783 \quad \text{over by } 0.542$$

$$3 \int_1^3 f(x) dx \approx \frac{0.5}{2} (8.01 + 2(6.02 + 4.44 + 3.8) + 3.27) \\ \approx 10.075$$

$$4 \begin{array}{c|c} y & 1.457 \quad \frac{\sqrt{10}}{4} \quad 2.225 \quad 2.358 \quad 2.489 \\ x & 2 \quad 2.25 \quad 2.5 \quad 2.75 \quad 3 \end{array}$$

$$\frac{1}{8} (1.457 + 2 \left(\frac{\sqrt{10}}{4} + 2.225 + 2.358 \right) + 2.489) \\ = 2.224$$

$$5 \frac{\pi}{2} (0.7011 + 2(0.732 + 0.8164) + 1) \\ = 1.257$$

$$6 \int_0^1 e^{-x^2}$$

$$\begin{array}{c|c} x & 0 \quad 0.5 \quad 1 \\ y & 1 \quad 0.7788 \quad 0.3679 \end{array} \quad \begin{array}{c|c} x & 0 \quad 0.25 \quad 0.5 \quad 0.75 \quad 1 \\ y & 1 \quad 0.9344 \quad 0.7788 \quad 0.5678 \quad 0.3679 \end{array}$$

$$\frac{0.5}{2} (1 + 2(0.7788) + 0.3679) \\ = 0.79375$$

$$\frac{0.25}{2} (1 + (0.9344 + 0.7788 + 0.5678) + 0.3679) \\ = 0.74249$$

$$\frac{d}{dx} e^{-x^2} = -2x e^{-x^2}$$

$$= -2x \cdot e^{-x^2} \cdot -2x + e^{-x^2} \cdot -2$$

$$= 4e^{-x^2} x^2 - 2e^{-x^2}$$

$$= e^{-x^2} (4x^2 - 2)$$

so the sign of $\frac{d}{dx} e^{-x^2}$ changes as x varies (so gradient changes)

so we cannot know if the estimate is over or under estimate.

$$7 \int_0^2 (1+x)^2 dx$$

$$h = \frac{1}{10}$$

$$\frac{1}{20} (1 + 2(1.1268 + 1.601 + 2.8126 + 0.936) + 14.551)$$

$$= 1.9779$$

$$ii) (1+x)^2 \approx 1 + 2x^2 + \frac{12 \cdot 11}{2!} x^4 + \frac{12 \cdot 11 \cdot 6}{3!} x^6 + \frac{12 \cdot 11 \cdot 10 \cdot 9}{4!} x^8$$

$$\approx 1 + 2x^2 + 66x^4 + 220x^6 + 495x^8$$

iii) Integrating:

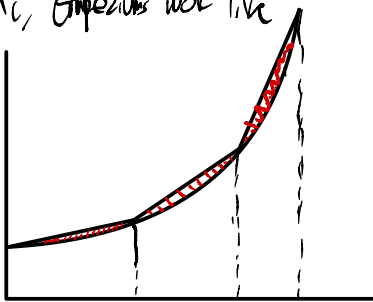
$$\left[55x^9 + \frac{220}{7}x^7 + \frac{66}{5}x^5 + 4x^3 + x \right]_0^{0.5}$$

$$= 55 \cdot 0.5^9 + \frac{220}{7} \cdot 0.5^7 + \frac{66}{5} \cdot 0.5^5 + 4 \cdot 0.5^3 + 0.5$$

$$= 1.7655$$

iv) iii, Trapeziums look like

So overestimate



ii) additional terms are missing
∴ underestimate.

Exam Questions

$$1) a) \frac{1}{2} (8 + 2(8.5 + 42 + 64) + 0.6)$$

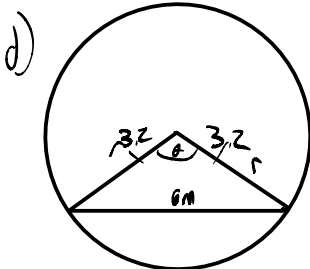
$$50.1 \text{ m}^2$$

$$ii) £ 24.55$$

b) estimation is quite accurate

gradient changes rapidly; Trapezium rule is more accurate on linear graphs

c) more strips.



$$\theta = 2 \arcsin \frac{3}{3.2}$$

$$\text{area} = 3.2 \cdot \theta - \frac{1}{2} \cdot 6 \cdot 3.2 \cos \theta$$

$$= 4.4377 \text{ m}^2$$

$$\therefore \text{cost} = £ 0.75$$

$$\frac{1}{2} r^2 \theta?$$

$$\begin{aligned}
 2a) \quad S &= h^2 \cdot h + 4h^2 \cdot h + 9h^2 \cdot h + \dots + n^2 \cdot h \\
 &= h(1^2 + 4h^2 + 9h^2 + \dots + (n)^2) \\
 &= h^3 \left(\sum_{k=1}^n k^2 \right) \\
 &= h^3 \cdot \left[\frac{1}{6} n(n+1)(2n+1) \right] \\
 &= \frac{1}{6} h^3 n(n+1)(2n+1)
 \end{aligned}$$

$$\begin{aligned}
 b) \text{ from a), } S &= \frac{1}{6} h^3 n(n+1)(2n+1) \quad n = \frac{p}{h} \\
 &= \frac{1}{6} h^3 \cdot \frac{p}{h} \left(\frac{p}{h} + 1 \right) \left(2\frac{p}{h} + 1 \right) \\
 &= \frac{1}{6} p h^2 \left(\frac{p}{h} + 1 \right) \left(2\frac{p}{h} + 1 \right) \\
 &= \frac{1}{6} p (p+h)(2p+h)
 \end{aligned}$$

$$\begin{aligned}
 c) \text{ As } h \rightarrow 0, \quad S &= \frac{1}{6} p(p+0)(2p+0) \\
 &= \frac{1}{6} \cdot p \cdot p \cdot 2p \\
 &= \underline{\underline{\frac{1}{3} p^3}} = A
 \end{aligned}$$